Speech Recognition

Exercises NLTK2 on Stemming, Lemmatizing and PoS Tagging

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Part 01

Create a wordlist with at least the following words:

```
[ 'cats', 'fish', 'runs', 'played', 'gotten', 'feed', 'shed', 'emptied', 'proceed',
    'generously', 'strictly', 'modernization']
```

Use the NLTK's Porter stemmer and one other stemmer to stem every word in the list. Are there differences in the results?

Find 3 examples where one stemmer returns a better result than the other.

Solution

```
# see https://www.nltk.org/howto/stem.html
from nltk.stem.porter import *
stemmer1 = PorterStemmer()
stemmer2 = LancasterStemmer()
wl = [ 'cats', 'fish', 'runs', 'played', 'gotten', 'feed', 'shed', 'emptied', 'proceed',
    'generously', 'strictly', 'modernization']
stems1 = [stemmer1.stem(word) for word in wl]
stems2 = [stemmer2.stem(word) for word in wl]
print("Stems from stemmer 1 - Porter:")
print(stems1)
# ['cat', 'fish', 'run', 'play', 'gotten', 'feed', 'shed', 'empti', 'proceed', 'gener',
    'strictli', 'modern']
print("Stems from stemmer 2 - Lancaster:")
print(stems2)
# ['cat', 'fish', 'run', 'play', 'got', 'fee', 'shed', 'empty', 'process', 'gen',
    'strictly', 'modern']
```

Part 02

Take words from the wordlist of the previous exercise and lemmatize the words. Compare the results with the results of stemming.

Find 3 examples where lemmatizer returns a better result than the Porter Stemmer.

Solution

```
from nltk.stem import WordNetLemmatizer
lemmatizer = WordNetLemmatizer()
wl = [ 'cats', 'fish', 'runs', 'played', 'gotten', 'feed', 'shed', 'emptied', 'proceed',
    'generously', 'strictly', 'modernization']
lems = [lemmatizer.lemmatize(word) for word in wl]
```

```
print(lems)
# ['cat', 'fish', 'run', 'played', 'gotten', 'feed', 'shed', 'emptied', 'proceed',
    'generously', 'strictly', 'modernization']
for w in wl:
    print(lemmatizer.lemmatize(w))
    print(lemmatizer.lemmatize(w, 'n'))
    print(lemmatizer.lemmatize(w, 'v'))
    print(lemmatizer.lemmatize(w, 'a'))
```

Part 03

Write programs to process the Brown Corpus and find answers to the following questions:

1) Which nouns are more common in their plural form, rather than their singular form? (Only consider regular plurals, formed with the -s suffix.)
2) Which word has the greatest number of distinct tags. What are they, and what do they represent?
3) List tags in order of decreasing frequency. What do the 20 most frequent tags represent?
4) Which tags are nouns most commonly found after? What do these tags represent?

Part 04

Generate some statistics for tagged data to answer the following questions:

1) What proportion of word types are always assigned the same part-of-speech tag?
2) How many words are ambiguous, in the sense that they appear with at least two tags?
3) What percentage of word tokens in the Brown Corpus involve these ambiguous words?

Practical

Part 01

Your chatbot should analyse the input. Apply a tagger to an input sentence and add the tags to the output in the chatbot. Start with the pos_tag() function. You may test another one later. Compare the results with the results of the grammar parsing from the former exercise.